

I am a software engineer devoted to the development of software systems and to the education and research on related subjects. During the last 3 years I have been leading the development of the new information system for DOSPU, the health insurance organization of the UNSL (National University of San Luis)¹, using technologies such as Angular, Spring and PostgreSQL. Before becoming a professor and researcher, I worked in the software industry for almost 8 years as developer, software architect and project leader. Sometimes I am regarded as a generalist in this context. I have over 20 years of teaching experience at universities and I currently have a professor appointment at UNSL. My main research interests are calculi to support the job of software architects, agile methodologies, project management, electronic governance and semantic interoperability. I had the privilege to be a researcher at United Nations University and part of research projects funded by the European Community. In my spare time I like reading, trail running (I am part of a team called “Las Cabras” – the goats) and swimming.

1. Formal education

- PhD in Computer Science – *ongoing*, UNSL, San Luis, Argentina (Arg.)
- Magister in Software Engineering – completed on 2008, UNSL, San Luis, Arg.
- Systems Engineer – completed on 2001, UNICEN², Tandil, Arg.

2. Languages

- Spanish (native)
- English
 - Academic IELTS 2008, general 7, writing 6, reading 8, speaking 7, listening 7.5
 - Online course “Writing in Sciences” Stanford University 2014, approved
- Portuguese (basic level informally acquired upon living 18 months in Portugal)

3. Teaching

3.1. Postgraduate

Professor in charge, UNSL, *Gestión de Proyectos de Software*, 80 hs, Maestría en Calidad del Software, 2024

Professor in charge, UTN - FRT³, *Modelado Conceptual de Sistemas de Información*, 60 hs., Especialización en Ingeniería en Sistemas de Información, Sep – Oct 2019, Apr – May 2022

Collaborating professor, UNSL, since 2022:

- *Arquitecturas de Sistemas de Software*, 80 hs, Maestría en Ingeniería de Software
- *Gestión de Proyectos de Software*, 80 hs, Maestría en Ingeniería de Software
- *Arquitecturas de Sistemas de Software*, 80 hs, Especialización en Ingeniería de Software
- *Análisis de Artefactos de Software*, 80 hs, Especialización en Ingeniería de Software

Collaborating professor, UNSL:

- *Arquitecturas de Software*, 60 hs, Doctorado en Ingeniería Informática, Jun – Jul 2019

¹<http://www.unsl.edu.ar>

²<http://www.unicen.edu.ar>

³<https://frt.utn.edu.ar/>

- *Tópicos Avanzados de Especificación y Verificación Formal de Software*, Maestría en Ingeniería en Software, Aug – Nov 2013

Collaborating professor, UTN - FRT, *Modelado Conceptual de Sistemas de Información*, 60 hs, Maestría en Ingeniería en Sistemas de Información, 60 hs., Aug – Sep 2017, Sep – Oct 2014, Jun 2013

Collaborating professor, UNSJ⁴, *Arquitecturas de Sistemas de Software y Tecnologías Asociadas*, 24 hs, Maestría en Informática, Oct – Nov 2013

Collaborating professor, UMinho⁵, Portugal, *Formal Methods in Software Engineering, Master in Informatics Engineering*, Sep 2011 – Jul 2012

Professor in charge, UNPSJB⁶, *Introducción a la Ingeniería de Ontologías*, 80 hs, Sep 2015 – Jul 2016

Professor in charge, UNCA⁷, *La Web Semántica*, 40 hs, Nov 2010 – Dec 2010

3.2. Bachelor degree

Professor in charge, UNSL, Ingeniería en Informática (Ing. Inf.):

- *Programación III*, since 2025
- *Sistemas de Tiempo Real*, since 2020

Professor in charge, UNPSJB, *Sistemas Distribuidos*, Licenciatura en Informática, 2014-2023

Collaborating professor, UNSL, Ing. Inf.:

- *Métodos Formales*, since 2021
- *Programación III*, 2020–2024
- *Ingeniería de Software I* since 2020

Collaborating professor, UNSL, Tecnicatura Universitaria Web (TUW), *Ingeniería Web*, 2020

Senior teaching assistant, UNSL:

- *Métodos Formales*, 2016 – 2019, Ing. Inf.
- *Programación III*, 2014 – 2019, Ing. Inf.
- *Ingeniería de Software I*, 2014, 2015, 2017, 2018, Licenciatura en Ciencias de la Computación (Lic. CC.) Profesorado en Ciencias de la Computación (Prof. CC.)
- *Ingeniería Web*, 2012 – 2018, TUW
- *Ingeniería de Software*, 2013 – 2014, TUW
- *Programación I*, 2013, Lic. CC., Prof. CC., Ing. Inf. & Ingeniería en Computación (Ing.Comp.)
- *Programación I*, 2012, Lic. CC., Prof. CC.
- *Programación III*, 2010, TUW
- *Ingeniería de Software*, 2010, TUW

Collaborating professor, UMinho, Portugal, *Processos e Concorrência*, Licenciatura em Ciências da Computação, 2011–2012

Teaching assistant, UNSL:

⁴<https://www.unsj.edu.ar/>

⁵<https://www.uminho.pt>

⁶<http://www.unp.edu.ar>

⁷<http://www.unca.edu.ar/>

- *Ingeniería Web*, 2010, TUW
- *Programación I*, 2009, Lic. CC.
- *Trabajo Final*, 2009, TUW
- *Programación III*, 2009, TUW

Teaching assistant, UNSL:

- *Ingeniería de Software II*, 2002, Lic. CC.
- *Desarrollo de Aplicaciones Web*, 2002
- *Planeamiento Informático*, 2002, Lic. CC.

Undergraduate teaching assistant, UNICEN, Ingeniería en sistemas:

- *Metodologías de Desarrollo de Software 1* 1998, 2001
- *Arquitecturas de Sistemas Interactivos*, 2001
- *Programación Orientada a Objetos*, 2000
- *Comunicación de Datos I*, 1998, 1999
- *Introducción a la Programación I*, 1996 – 1998, 2001
- *Introducción a la Programación II*, 1996 – 1999

3.3. Extracurricular and professional development courses

Professor in charge, UNSL, extracurricular course, *L^AT_EX en acción*, 16 hs, Oct – Dec 2020

Professor in charge, UNICEN, CACIC 2018 – school course, *Una introducción al desarrollo de ontologías OWL*, 15 hs, Oct 2018

Professor in charge, UTN – Facultad Regional Córdoba, professional development course, *Arquitecturas de software como sistemas reactivos: modelado, especificación y verificación*, 20 hs, Sep – Oct 2016.

4. Software development

- Project leader & developer (dev.), SI-DOSPU | since 11/2020 | UNSL, San Luis (SL), Arg.
 - SI-DOSPU is the new Information System (IS) that will replace the outdated information system of DOSPU, the health insurance organization for UNSL's employees. The project aims at developing an agile and affiliated-centred system in three phases: replace the obsolete IS with a flexible and web-enabled IS, capable of adapting to an increasingly evolving and challenging context in Argentina; provide decision-support information to senior managers; and re-engineer processes to offer affiliates a personalized service that includes prevention and anticipated detection of pathologies. The development is carried out applying a number of agile methodologies' practices and cutting edge technologies (see third item for more detail).
 - My main role in the development of SI-DOSPU is leading a dynamic team of advanced UNSL's students of informatics degrees. This includes selecting, training and managing the team. I am also in charge of defining the development process, leading business processes and system analysis, eliciting requirements, architectural design, database (and infrastructure, in general) design and administration, and training end-users.
 - project management, architectural patterns, design patterns, scrum, refactoring, TDD, BDD, CI, evolutionary database design, REST services, OOP, UML | Java, Spring, Angular, TypeScript, PostgreSQL, Jira, Linux, Kubernetes, Docker, Flyway, Maven, Cucumber, Jenkins, jakarta persistence
- Project leader & dev., Moambue | since 10/2023 | UNSL, SL-Arg.

- The project aims at providing affordable, agile and citizen-centred information systems to health insurance organizations of national universities. It received funding for developing a proof of concept.
- I am leading this project, which I conceived upon the experience I have gained in the development of SI-DOSPU.
- Project management, Java, PostgreSQL, Apache Beam
- Researcher & dev., Archery translator | 6 - 12/2011 | UMinho, Braga, Portugal
 - Archery is an architectural description language for modelling, animating, analysing and verifying software architectures. It emerged as an outcome of my research towards supporting software architects jobs and it is in an experimental stage.
 - I formulated the language and implemented the translator to the mCRL2 language.
 - Java, Antlr, mCRL2
- Researcher & dev., Semantic Interoperability for Programmable Messaging Middleware | 1/2007 - 6/2008 | United Nations University - International Institute for Software Technology, Macau, China
 - A component for the Government-Enterprise Ecosystem Gateway (G-EEG) providing ontology-based semantic interoperability. G-EEG is a middleware enabling programmable messaging developed for the collaborative delivery of public services.
 - I researched and developed the semantic interoperability component.
 - Java, Description Logics, OWL-DL, Pellet
- Team Leader, Aptela | 1/2006 - 12/2006 | Novamens S.A, SL-Arg.
 - A VoIP hosted PBX and video-communication services. It was one of the first providers of which could be regarded nowadays as Voice as a Service product. In 2011 it was acquired by Vonage Holdings Corp. (which in turn was acquired by Ericsson in 2021) a leading provider of communications services in USA.
 - I led and coached a team of under graduated developers.
 - Java, JEE, JBoss, SOAP, REST Services, Eclipse, Ant, JUnit, and PostgreSQL
- Team leader and dev., KBee | 7/2005 - 12/2005 | Novamens S.A, SL-Arg.
 - A sophisticated content manager customized for US Real State Monitoring.
 - I led and coached a team of junior under graduated developers.
 - Java, JEE, Cocoon, Eclipse, JUnit and PostgreSQL
- Team leader and dev., Waitless | 1/2005 - 6/2005 | Novamens S.A, SL-Arg.
 - A web app. for restaurants enabling the reception and processing of on line orders.
 - I led and coached a team of junior under graduated developers performing minor development tasks in the project.
 - Java, JEE, Apache Tomcat, Struts, Hibernate, Velocity, Eclipse, JUnit and PostgreSQL
- Software architect & dev., ACU Smart | 06/2004 - 12/2004 | NEC Argentina S.A., SL-Arg.
 - A fingerprint access control and identification device.
 - I migrated its experimental ~5000 lines of JavaScript code, programmed into a single file, to a maintainable (with a logical design) Java-based production level solution.
 - Refactoring, architectural patterns, design patterns | Java, Eclipse
- Software architect & dev., San Luis Province Information Highway | 02/2003 – 06/2004 | NEC Argentina S.A., SL-Arg.
 - An e-Gov Web information system for San Luis province.
 - I introduced CI and promoted TDD. CI saved days of build efforts in every release. I was involved in the Public Security Suite and Online Employment Systems.
 - CI, TDD, refactoring, architectural patterns, design patterns | Enterprise Architect, Erwin, UML, RUP, .NET Environment, NUnit, NDoc, NAnt, CruiseControl.Net, Microsoft Business Integrator, XML/XSL/HTML, JavaScript, IIS web server, SQL Ser-

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- Details on soft. dev. appointments previous to 2004 will be provided upon request.

5. Research

- Agile methodologies and project management, since 2020. Seeks to understand to what extent agile management practices are suited for public sector organizations, particularly in the domain of argentinean health insurance organizations for universities, using the experience gained during the development of SI-DOSPU.
- Calculi to support the job of software architects, since 2011. Seeks adapting algebraic and logic formalisms (and their associated toolkits) to provide better tools for software architects dealing with distributed, heterogeneous and loosely coupled processes. Part of the research was developed in the context of research projects funded by the European Community and at the University of Minho, Portugal. The outcome includes two articles in journals and twelve conference papers.
- Electronic governance and semantic interoperability. Seeks understanding to what extent technologies based on ontology languages such as OWL (and description logics) could be used to provide semantic interoperability tools at a higher abstraction level in the context of public organizations. Part of the research was developed at the United Nations University, in Macau, China. The outcome includes one article in a journal and nine conference papers.